

Pathology-Level ML Model Performance Summary (Test Set)
Classical ML models trained on neural network pathology predictions

Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC Std	AUC 95% CI	F1 Mean	F1 Std	Samples	Positive Rate	Cross-Val Folds
Attention Pool Extra3	A Lines	Random Forest	0.758	0.026	[0.726-0.790]	0.665	0.041	43689	0.472	5
Attention Pool Extra3	A Lines	Logistic Regression	0.795	0.014	[0.778-0.813]	0.709	0.034	43689	0.472	5
Attention Pool Extra3	A Lines	XGBoost	0.794	0.013	[0.779-0.810]	0.694	0.029	43689	0.472	5
Attention Pool Extra3	Large Consolidation	Random Forest	0.863	0.013	[0.847-0.880]	0.479	0.078	43689	0.110	5
Attention Pool Extra3	Large Consolidation	Logistic Regression	0.919	0.004	[0.913-0.924]	0.554	0.039	43689	0.110	5
Attention Pool Extra3	Large Consolidation	XGBoost	0.901	0.012	[0.886-0.915]	0.553	0.064	43689	0.110	5
Attention Pool Extra3	Pleural Effusion	Random Forest	0.674	0.058	[0.602-0.745]	0.049	0.040	43689	0.011	5
Attention Pool Extra3	Pleural Effusion	Logistic Regression	0.808	0.039	[0.759-0.856]	0.063	0.039	43689	0.011	5
Attention Pool Extra3	Pleural Effusion	XGBoost	0.829	0.041	[0.778-0.880]	0.043	0.040	43689	0.011	5
Attention Pool Extra3	Other Pathology	Random Forest	0.644	0.024	[0.614-0.674]	0.492	0.062	43689	0.405	5
Attention Pool Extra3	Other Pathology	Logistic Regression	0.690	0.022	[0.662-0.717]	0.583	0.046	43689	0.405	5
Attention Pool Extra3	Other Pathology	XGBoost	0.684	0.018	[0.662-0.707]	0.513	0.066	43689	0.405	5
Mean Pool Extra3	A Lines	Random Forest	0.754	0.064	[0.675-0.833]	0.666	0.060	43689	0.472	5
Mean Pool Extra3	A Lines	Logistic Regression	0.790	0.036	[0.745-0.835]	0.719	0.042	43689	0.472	5
Mean Pool Extra3	A Lines	XGBoost	0.790	0.054	[0.723-0.856]	0.690	0.056	43689	0.472	5
Mean Pool Extra3	Large Consolidation	Random Forest	0.829	0.114	[0.687-0.971]	0.388	0.157	43689	0.110	5
Mean Pool Extra3	Large Consolidation	Logistic Regression	0.908	0.030	[0.872-0.945]	0.498	0.115	43689	0.110	5
Mean Pool Extra3	Large Consolidation	XGBoost	0.862	0.117	[0.716-1.007]	0.408	0.167	43689	0.110	5
Mean Pool Extra3	Pleural Effusion	Random Forest	0.678	0.064	[0.599-0.757]	0.068	0.055	43689	0.011	5
Mean Pool Extra3	Pleural Effusion	Logistic Regression	0.730	0.065	[0.648-0.811]	0.048	0.019	43689	0.011	5
Mean Pool Extra3	Pleural Effusion	XGBoost	0.819	0.066	[0.737-0.900]	0.048	0.052	43689	0.011	5
Mean Pool Extra3	Other Pathology	Random Forest	0.645	0.032	[0.606-0.685]	0.487	0.089	43689	0.405	5
Mean Pool Extra3	Other Pathology	Logistic Regression	0.672	0.031	[0.634-0.711]	0.489	0.274	43689	0.405	5
Mean Pool Extra3	Other Pathology	XGBoost	0.686	0.046	[0.629-0.743]	0.487	0.135	43689	0.405	5
Uniform Extra3	A Lines	Random Forest	0.753	0.033	[0.713-0.794]	0.661	0.026	43689	0.472	5
Uniform Extra3	A Lines	Logistic Regression	0.792	0.011	[0.778-0.806]	0.724	0.006	43689	0.472	5
Uniform Extra3	A Lines	XGBoost	0.795	0.017	[0.773-0.816]	0.698	0.014	43689	0.472	5
Uniform Extra3	Large Consolidation	Random Forest	0.868	0.027	[0.834-0.901]	0.421	0.106	43689	0.110	5
Uniform Extra3	Large Consolidation	Logistic Regression	0.912	0.009	[0.901-0.922]	0.530	0.013	43689	0.110	5
Uniform Extra3	Large Consolidation	XGBoost	0.902	0.016	[0.882-0.922]	0.449	0.117	43689	0.110	5
Uniform Extra3	Pleural Effusion	Random Forest	0.618	0.057	[0.547-0.689]	0.060	0.028	43689	0.011	5
Uniform Extra3	Pleural Effusion	Logistic Regression	0.794	0.016	[0.774-0.813]	0.053	0.017	43689	0.011	5
Uniform Extra3	Pleural Effusion	XGBoost	0.817	0.081	[0.716-0.917]	0.041	0.034	43689	0.011	5
Uniform Extra3	Other Pathology	Random Forest	0.635	0.019	[0.611-0.659]	0.490	0.046	43689	0.405	5
Uniform Extra3	Other Pathology	Logistic Regression	0.674	0.019	[0.651-0.698]	0.568	0.043	43689	0.405	5
Uniform Extra3	Other Pathology	XGBoost	0.677	0.016	[0.658-0.696]	0.499	0.063	43689	0.405	5